

Sertaç Akalın

sertacakalin0@gmail.com | +90 543 542 64 98
github.com/sertacakalin | linkedin.com/in/sertacakalin

Summary

Computer Engineering (English) graduate. I work on backend development using Java/Spring Boot and Python/FastAPI, with hands-on experience in REST API development, database operations, and service integrations. I aim to develop my technical skills through real-world projects in backend-focused roles while adapting to corporate software development processes.

Technical Skills

Backend: Java, Spring Boot, Spring Data JPA, Python, FastAPI

Databases: PostgreSQL, MySQL

AI: LangChain, RAG, OpenCV, YOLOv8

Tools: Git, Docker, Postman, Firebase

Education

Istanbul Arel University — B.Sc. in Computer Engineering (English)	2022–2026
Associate Degree in Web Design and Coding	2020–2022
Akdeniz University — Surveying and Mapping	2018–2020

Experience

Java Backend Developer Intern — CodeFirst 4 months

- Worked on backend development for a Spring Boot-based system with a microservice and layered (Controller–Service–Repository) architecture, including RESTful APIs and external service integrations
- Took part in database operations using Spring Data JPA and Hibernate ORM; used DTOs for data transfer between layers
- Took part in transaction management and JWT-based authentication, contributing to data consistency and security
- Used Postman/Swagger for testing and documentation, and Git/Docker for version control and environment management; also involved in logging

Software Engineering Intern — FEV Türkiye 6 months

- Took part in backend development with Python and FastAPI on SOVD (Service-Oriented Vehicle Diagnostics), an external project for the client Jaguar Land Rover / Tata
- Worked on REST-based diagnostic services and contributed to developing service endpoints with FastAPI
- Worked within a cross-cultural team of engineers from Germany and India; carried out assigned tasks and contributed to various solutions from an engineering perspective
- Took part in daily stand-up meetings and corporate workflows, gaining first-hand experience of teamwork and enterprise software development
- Developed my skills in problem analysis, solution design, and communication within an international team in a real engineering environment

Projects

Palmystra — AI-Powered Mobile App

FastAPI, LangChain, Computer Vision, SwiftUI

- Built a full-stack iOS application that combines computer vision and AI to generate personalized readings from a user's palm photo
- Developed an AI pipeline that detects palm lines through image processing and generates personalized readings via a LangChain-based knowledge base integration
- Built the backend with API security, rate limiting, and a test suite, and deployed it using Docker and Google Cloud Run

Hatched Area Violation Detection

YOLOv8, ByteTrack, OpenCV, Shapely, EasyOCR

- Built a computer vision system that detects vehicles entering hatched road areas in traffic videos
- Fine-tuned a YOLOv8 model on a self-collected Istanbul traffic dataset, achieving 85%+ detection accuracy, and performed multi-vehicle tracking with ByteTrack
- Added a pipeline of plate detection, OCR, and format validation to read Turkish license plates at the moment of a violation
- Designed detection, tracking, and violation analysis as separate modules, making the system more maintainable

Other

Languages: English (B2)

Military Service: Completed